

TrES-5b

R-0787

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_221027 · created 2026-07-28T06:51:12+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459879.6233 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.054
Transit depth (Rp/Rs)^2	3.21 +/- 1.93 [%]
Orbital Inclination (inc)	88.87 +/- 2.7
Airmass coefficient 1 (a1)	0.85 +/- 0.03
Airmass coefficient 2 (a2)	0.15 +/- 0.13
Scatter in the residuals of the lightcurve fit is	3.67 %
Best Comparison Star	#3 - [26, 240]
Optimal Method	PSF photometry
Transit Duration (day)	0.0852 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.179 ± 0.054 vs 0.142 (deviation 0.69σ)
Duration fit vs published	0.0852 d vs 0.0758 d (deviation 1.44σ)
Mid-time O–C	8.6 min
Depth SNR	3.3
Residual scatter	3.67 %

Light curve

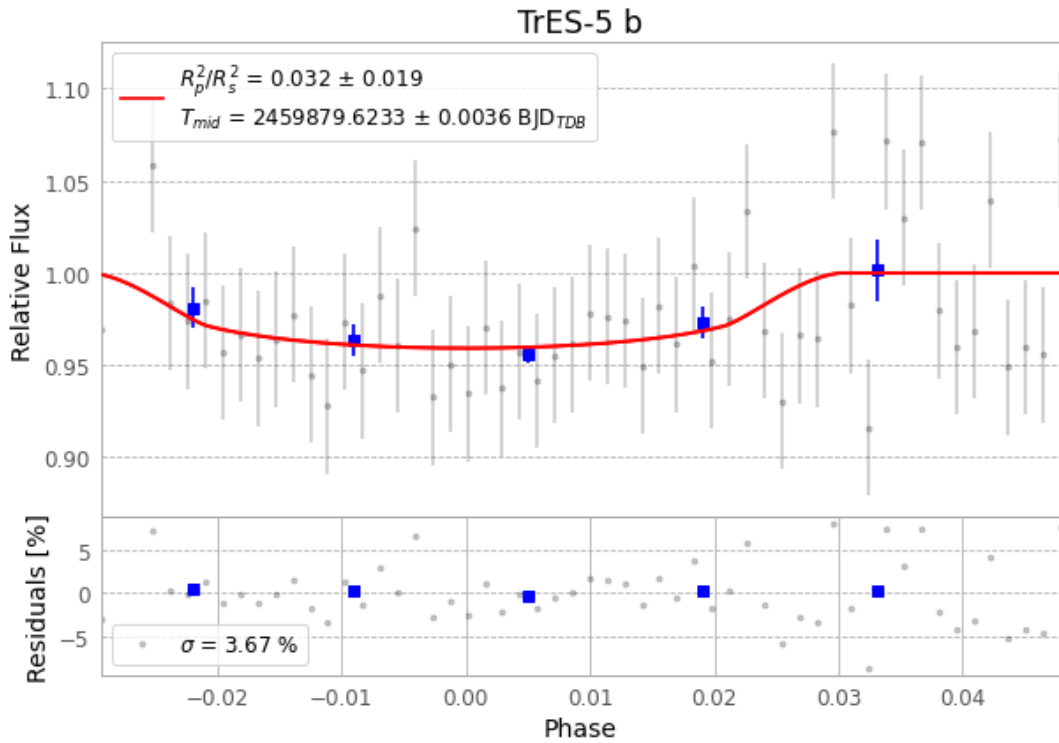


Figure 1 — FinalLightCurve_TrES-5 b_2022-10-26.png (SHA-256 f4d31442fe794225...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [431, 171], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9a959d032fbee017...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

54 FITS frames (35 MB), TRES-5221027015113.FITS → TRES-5221027043615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-221027.log (45d9e196f518...), FinalLightCurve_TrES-5 b_2022-10-26.png (f4d31442fe79...), FinalLightCurve_TrES-5 b_2022-10-26.csv (9fd6c68c295e...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 06:51Z.